1. Consider the database shown in Figure 1.2, write appropriate SQL DDL statements to create the database shown in Figure 1.2 in your MySQL account.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

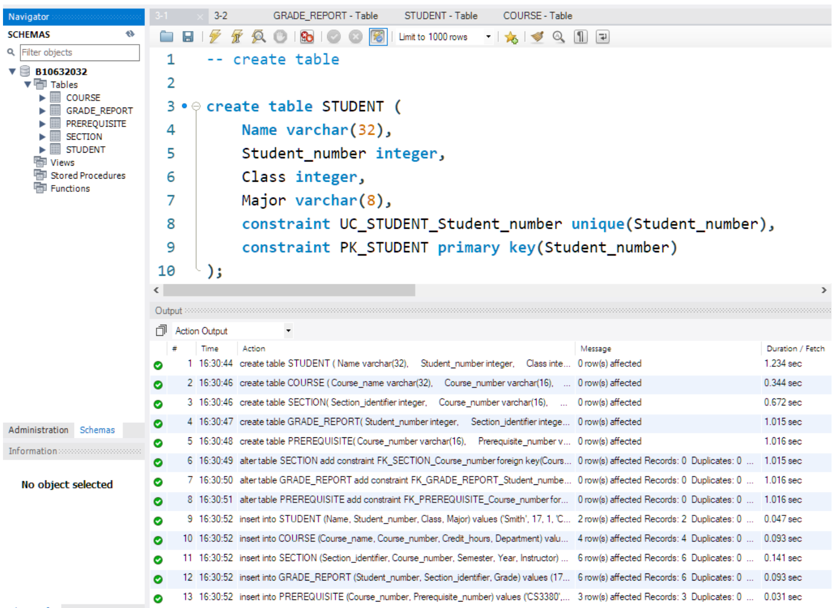
Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

```
-- create table


create table STUDENT (
    Name varchar(32),
    Student_number integer,
    Class integer,
    Major varchar(8),
    constraint UC_STUDENT_Student_number unique(Student_number),
    constraint PK_STUDENT primary key(Student_number)
);
```

```
create table COURSE (
   Course_name varchar(32),
    Course_number varchar(16),
    Credit_hours integer,
    Department varchar(8),
    constraint UC_COURSE_Course_number unique(Course_number),
    constraint PK_COURSE primary key(Course_number)
);

create table SECTION(
   Section_identifier integer,
    Course_number varchar(16),
    Semester varchar(8),
    Year char(2),
    Instructor varchar(32),
    constraint PK_SECTION primary key(Section_identifier)
);

create table GRADE_REPORT(
   Student_number integer,
    Section_identifier integer,
    Grade varchar(1),
    constraint PK_GRADE_REPORT primary key(Section_identifier, Student_number)
);

create table PREREQUISITE(
   Course_number varchar(16),
    Prerequisite_number varchar(16),
    constraint PK_PREREQUISITE primary key(Course_number, Prerequisite_number)
);
```

```
-- FK

alter table SECTION
    add constraint FK_SECTION_Course_number foreign key(Course_number) references COURSE(Course_number);

alter table GRADE_REPORT
    add constraint FK_GRADE_REPORT_Student_number foreign key(Student_number) references STUDENT(Student_number),
    add constraint FK_GRADE_REPORT_Section_identifier foreign key(Section_identifier) references SECTION(Section_identifier);

alter table PREREQUISITE
    add constraint FK_PREREQUISITE_Course_number foreign key(Course_number) references COURSE(Course_number),
    add constraint FK_PREREQUISITE_Prerequisite_number foreign key(Prerequisite_number) references COURSE(Course_number);


-- insert tuple

insert into STUDENT (Name, Student_number, Class, Major)
values ('Smith', 17, 1, 'CS'),
       ('Brown', 8, 2, 'CS');

insert into COURSE (Course_name, Course_number, Credit_hours, Department)
values ('Intro to Computer', 'CS1310', 4, 'CS'),
       ('Data Structures', 'CS3320', 4, 'CS'),
       ('Discrete Mathematics', 'MATH2410', 3, 'MATH'),
       ('Database', 'CS3380', 3, 'CS');
```

```
insert into SECTION (Section_identifier, Course_number, Semester, Year, Instructor)
values (85, 'MATH2410', 'Fall', '07', 'King'),
       (92, 'CS1310', 'Fall', '07', 'Anderson'),
       (102, 'CS3320', 'Spring', '08', 'Knuth'),
       (112, 'MATH2410', 'Fall', '08', 'Chang'),
       (119, 'CS1310', 'Fall', '08', 'Anderson'),
       (135, 'CS3380', 'Fall', '08', 'Stone');

insert into GRADE_REPORT (Student_number, Section_identifier, Grade)
values (17,112,'B'),
       (17,119,'C'),
       (8,85,'A'),
       (8,92,'A'),
       (8,102,'B'),
       (8,135,'A');

insert into PREREQUISITE (Course_number, Prerequisite_number)
values ('CS3380', 'CS3320'),
       ('CS3380', 'MATH2410'),
       ('CS3320', 'CS1310');
```



Consider the database shown in Figure 5.6, whose schema is shown in Figure 5.7. Write appropriate SQL DDL statements to create the database shown in Figure 5.6 in your MySQL account.

Figure 5.6
One possible database state for the COMPANY relational database schema.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

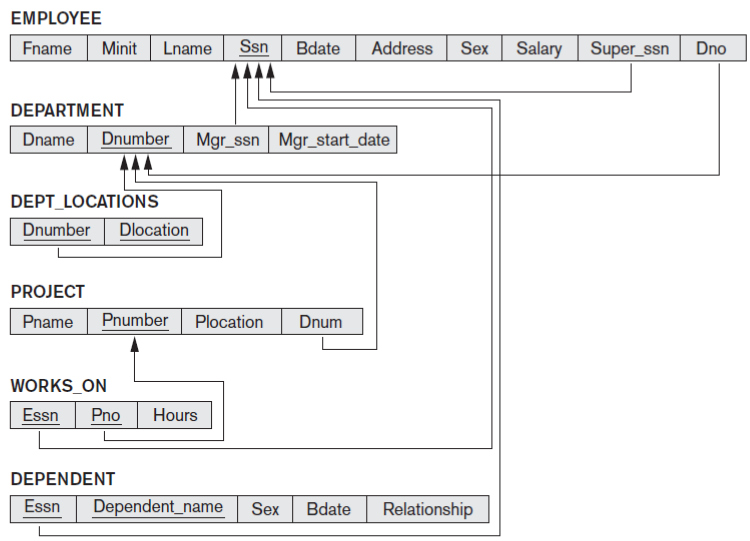
PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Figure 5.7
Referential integrity constraints displayed on the COMPANY relational database schema.



-- create table

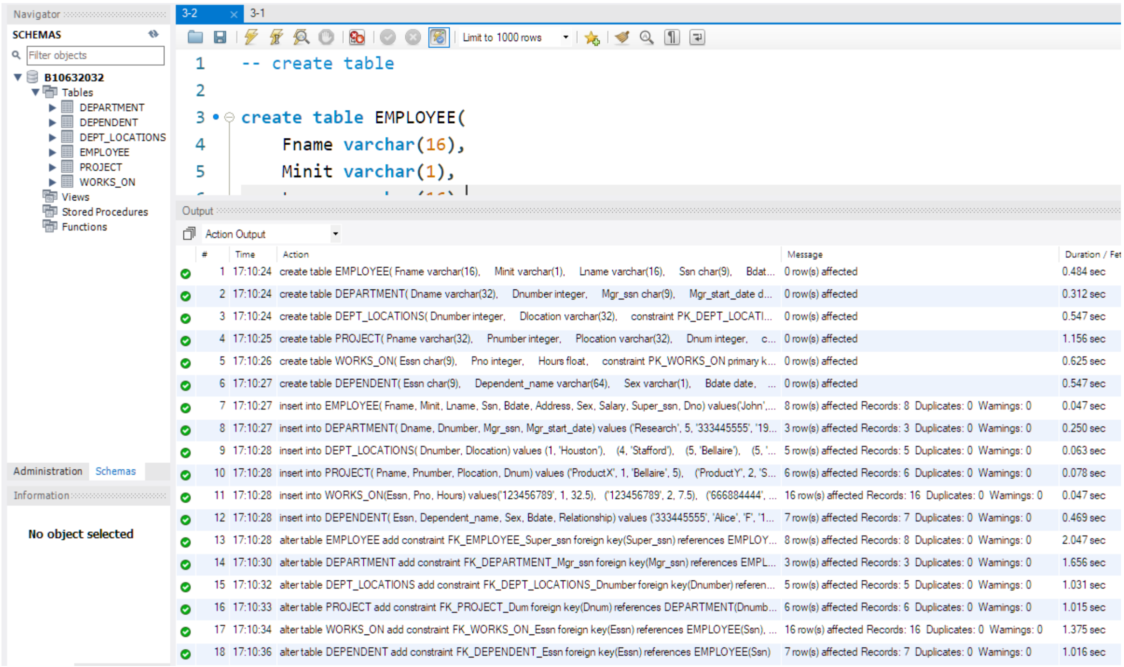
```
create table EMPLOYEE(
   Fname varchar(16),
    Minit varchar(1),
    Lname varchar(16),
    Ssn char(9),
    Bdate date,
    Address varchar(128),
    Sex varchar(1),
    Salary integer,
    Super_ssn char(9),
    Dno integer,
    constraint UC_EMPLOYEE unique(Ssn),
    constraint PK_EMPLOYEE primary key(Ssn)
);

create table DEPARTMENT(
   Dname varchar(32),
    Dnumber integer,
    Mgr_ssn char(9),
    Mgr_start_date date,
    constraint UC_DEPARTMENT unique(Dnumber),
    constraint PK_DEPARTMENT primary key(Dnumber)
);

create table DEPT_LOCATIONS(
   Dnumber integer,
    Dlocation varchar(32),
    constraint PK_DEPT_LOCATIONS primary key(Dnumber, Dlocation)
);
```

```
create table PROJECT(
   Pname varchar(32),
    Pnumber integer,
    Plocation varchar(32),
    Dnum integer,
    constraint UC_PROJECT unique(Pnumber),
    constraint PK_PROJECT primary key(Pnumber)
);

create table WORKS_ON(
   Essn char(9),
    Pno integer,
    Hours float,
    constraint PK_WORKS_ON primary key (Essn, Pno)
);

create table DEPENDENT(
   Essn char(9),
    Dependent_name varchar(64),
    Sex varchar(1),
    Bdate date,
    Relationship varchar(16),
    constraint PK_DEPENDENT primary key(Essn, Dependent_name)
);


-- insert data

insert into EMPLOYEE( Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary,
Super_ssn, Dno)
```

```
values('John', 'B', 'Smith', '123456789', '1965-01-09', '731 Fondren, Houston, TX',
'M', 30000, '333445555', 5),
      ('Franklin', 'T', 'Wong', '333445555', '1955-12-08', '638 Voss, Houston, TX',
'M', 40000, '888665555', 5),
      ('Alicia', 'J', 'Zelaya', '999887777', '1968-01-19', '3321 Castle, Spring,
TX', 'F', 25000, '987654321', 4),
      ('Jennifer', 'S', 'Wallace', '987654321', '1941-06-20', '291 Berry, Bellaire,
TX', 'F', 43000, '888665555', 4),
      ('Ramesh', 'K', 'Narayan', '666884444', '1962-09-15', '975 Fire Oak, Humble,
TX', 'M', 38000, '333445555', 5),
      ('Joyce', 'A', 'English', '453453453', '1972-07-31', '5631 Rice, Houston,
TX', 'F', 25000, '333445555', 5),
      ('Ahmad', 'V', 'Jabbar', '987987987', '1969-03-29', '980 Dallas, Houston,
TX', 'M', 25000, '987654321', 4),
      ('James', 'E', 'Borg', '888665555', '1937-11-10', '450 Stone, Houston, TX',
'M', 55000, null, 1);

insert into DEPARTMENT( Dname, Dnumber, Mgr_ssn, Mgr_start_date)
values ('Research', 5, '333445555', '1988-05-22'),
       ('Administration', 4, '987654321', '1995-01-01'),
       ('Headquarters', 1, '888665555', '1981-06-19');

insert into DEPT_LOCATIONS( Dnumber, Dlocation)
values (1, 'Houston'),
       (4, 'Stafford'),
       (5, 'Bellaire'),
       (5, 'Sugarland'),
       (5, 'Houston');

insert into PROJECT( Pname, Pnumber, Plocation, Dnum)
values ('ProductX', 1, 'Bellaire', 5),
```

```
        ('ProductY', 2, 'Sugarland', 5),
        ('ProductZ', 3, 'Houston', 5),
        ('Computerization', 10, 'Stafford', 4),
        ('Reorganization', 20, 'Houston', 1),
        ('Newbenefits', 30, 'Stafford', 4);

insert into WORKS_ON(Essn, Pno, Hours)
values('123456789', 1, 32.5),
      ('123456789', 2, 7.5),
      ('666884444', 3, 40),
      ('453453453', 1, 20),
      ('453453453', 2, 20),
      ('333445555', 2, 10),
      ('333445555', 3, 10),
      ('333445555', 10, 10),
      ('333445555', 20, 10),
      ('999887777', 30, 30),
      ('999887777', 10, 10),
      ('987987987', 10, 35),
      ('987987987', 30, 5),
      ('987654321', 30, 20),
      ('987654321', 20, 15),
      ('888665555', 20, null);

insert into DEPENDENT( Essn, Dependent_name, Sex, Bdate, Relationship)
values ('333445555', 'Alice', 'F', '1986-04-05', 'Daughter'),
        ('333445555', 'Theodore', 'M', '1983-10-15', 'Son'),
        ('333445555', 'Joy', 'F', '1958-05-03', 'Spouse'),
        ('987654321', 'Abner', 'M', '1942-02-28', 'Spouse'),
        ('123456789', 'Michael', 'M', '1988-01-04', 'Son'),
        ('123456789', 'Alice', 'F', '1988-12-30', 'Daughter'),
```

```
      ('123456789', 'Elizabeth', 'F', '1967-05-05', 'Spouse');



-- FK

alter table EMPLOYEE
    add constraint FK_EMPLOYEE_Super_ssn foreign key(Super_ssn) references EMPLOYEE(Ssn),
    add constraint FK_EMPLOYEE_Dno foreign key(Dno) references DEPARTMENT(Dnumber);

alter table DEPARTMENT
    add constraint FK_DEPARTMENT_Mgr_ssn foreign key(Mgr_ssn) references EMPLOYEE(Ssn);

alter table DEPT_LOCATIONS
    add constraint FK_DEPT_LOCATIONS_Dnumber foreign key(Dnumber) references DEPARTMENT(Dnumber);

alter table PROJECT
    add constraint FK_PROJECT_Dum foreign key(Dnum) references DEPARTMENT(Dnumber);

alter table WORKS_ON
    add constraint FK_WORKS_ON_Essn foreign key(Essn) references EMPLOYEE(Ssn),
    add constraint FK_WORKS_ON_Pno foreign key(Pno) references PROJECT(Pnumber);

alter table DEPENDENT
    add constraint FK_DEPENDENT_Essn foreign key(Essn) references EMPLOYEE(Ssn);
```



6.10. Specify the following queries in SQL on the COMPANY relational database schema shown in Figure 5.5. Show the result of each query if it is applied to the COMPANY database in Figure 5.6.

a. Retrieve the names of all employees in department 5 who earns more than 3000 and works on ProductZ project.

b. List the names of all employees who are from Houston, Texas and works under manager 333445555.

c. Find the names of all employees who are working in the project Computerization.

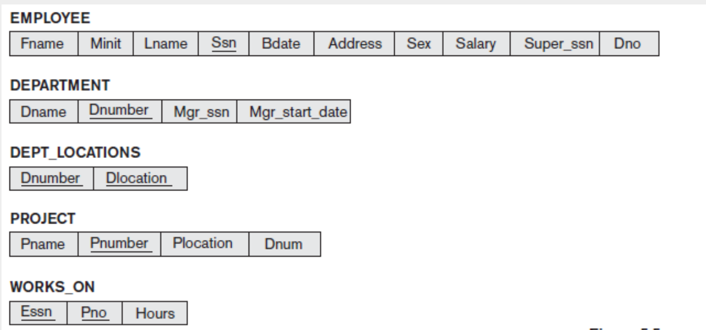


Figure 5.5
Schema diagram fo
COMPANY relation
database schema.

a. select Fname, Lname

from EMPLOYEE E, PROJECT P, WORKS_ON W

where E.Dno = 5 andE.Salary > 3000 andP.Pname = 'ProductZ' and P.Pnumber = W.Pno

and W.Essn = E.Ssn;

Fname	Lname

Franklin	Wong
Ramesh	Narayan

b. select Fname, Lname

from EMPLOYEE

where Address like '%Houston%TX%' and Super_ssn = '333445555';

Fname	Lname
John	Smith
Joyce	English

c. select Fname, Lname

from EMPLOYEE E, PROJECT P, WORKS_ON W

where P.Pname = 'Computerization' and P.Pnumber = W.Pno and W.Essn = E.ssn;

Fname	Lname
Franklin	Wong
Ahmad	Jabbar
Alicia	Zelaya

6.11. Specify the updates of Exercise 5.11 using the SQL update commands.

a. Insert <'Sophia', 'M', 'Wood'. '973442298'. '1974-05-21', '23 S Lamar Blvd. Rd, Austin, TX', 'F', 62000, '222445555', 5> into EMPLOYEE.

b. Insert <'6Sigma', 4, 'Austin', 4> into PROJECT.

c. Insert <'Information Technology', 2, '987987987', '2007-10-01' > into DEPARTMENT.

d. Insert <'777624972', 15, '40.0'> into WORKS_ON.

e. Insert <'888665555', 'John', 'M', null, 'Son'> into DEPENDENT.

f. Delete the DEPENDENT tuples with Essn = '987654321'.

g. Delete the DEPARTMENT tuples with Dnumber = 5.

h. Delete the WORKS_ON tuples with Pno = 30.

i. Modify the Plocation and Dnum of the PROJECT tuples with Dnum = 5 to 'Houston' and 1, respectively.

j. Modify the Super_ssn attribute of the EMPLOYEE tuple with Ssn = '888665555' to null. k. Modify the Pnumber attribute of the PROJECT tuple with Pnumber = 30 to 40

a. insert into EMPLOYEE(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)
values('Sophia', 'M' , 'Wood'. '973442298'. '1974-05-21', '23 S Lamar Blvd. Rd, Austin, TX', 'F' , 62000, '222445555', 5);
b. insert into PROJECT(Pname, Pnumber, Plocation, Dnum)
values ('6Sigma', 4, 'Austin', 4);
c. insert into DEPARTMENT(Dname, Dnumber, Mgr_ssn, Mgr_start_date)
values ('Information Technology', 2, '987987987', '2007-10-01');
d. insert into WORKS_ON(Essn, Pno, Hours)
values('777624972', 15, '40.0');
e. insert into DEPENDENT(Essn, Dependent_name, Sex, Bdate, Relationship)
values ('888665555', 'John', 'M' , null, 'Son');
f. delete from DEPENDENT
where Essn = '987654321' ;
g. delete from DEPARTMENT
where Dnumber = 5;
h. delete from WORKS_ON
where Pno = 30;
i. update PROJECT set Plocation = 'Houston', Dnum = 1 where Dnum = 5;
j. update EMPLOYEE set Super_ssn = null where Ssn = '888665555';
k. update PROJECT set Pnumber = 40 where Pnumber = 30;

6.12. Specify the following queries in SQL on the database schema of Figure 1.2. a. Retrieve the course names of all the courses that comes under the department of 'cs' (computer science). b. Retrieve the names of all courses along with the name of the instructor taught during the fall of 2008.

c. For each section taught by Professor Anderson, retrieve the course number, semester, year, and number of students who took the section.

d. Retrieve the name and transcript of each junior student (Class = 1) majoring in mathematics (MATH). A transcript includes course name, course number, credit hours, semester, year, and grade for each course completed by the student.

a. select Course_name
from COURSE
where upper(Department) like upper('cs');

Course_name
Intro to Computer
Database
Data Structures

b. select Course_name, Instructor
from COURSE C, SECTION S
where S.Semester = 'fall' and Year = '08' and S.Course_number = C.Course_number;

Course_name	Instructor
Discrete Mathematics	Chang
Intro to Computer	Anderson
Database	Stone

c. select Course_number, Semester, Year, ST.Student_number
from SECTION SE, STUDENT ST, GRADE_REPORT G
where SE.Instructor = 'Anderson' and ST.Student_number = G.Student_number and SE.Section_identifier = G.Section_identifier;

Course_number	Semester	Year	Student_number
CS1310	Fall	07	8
CS1310	Fall	08	17

d. select ST.Student_number, Course_name, C.Course_number, Credit_hours, Semester, Year, Grade
from COURSE C, SECTION SE, STUDENT ST, GRADE_REPORT G
where ST.Class = 1 and ST.Major = 'MATH';

Student_number	Course_name	Course_number	Credit_hours	Semester	Year	Grade

6.13. Write SQL update statements to do the following on the database schema shown in Figure 1.2.

a. Insert a new course, <'Financial Accounting', 'fac4390', 5, 'BUSINESS'>.

b. Insert a new section, < 145, 'fac4390', 'Fall', '17', 'Hanif' >.

c. Insert a new student, <'Robin', 34, 3, 'BUSINESS'>.

d. Update the record for the student whose student number is 17 and change his class from 1 to 3.

a. insert into COURSE (Course_name, Course_number, Credit_hours, Department)
values ('Financial Accounting', 'fac4390', 5, 'BUSINESS');

b. insert into SECTION (Section_identifier, Course_number, Semester, Year, Instructor)
values (145, 'fac4390', 'Fall', '17', 'Hanif');

c. insert into STUDENT (Name, Student_number, Class, Major)
values ('Robin', 34, 3, 'BUSINESS');

d. update STUDENT set Class = 3 where Student_number = 17;

7.5. Specify the following queries on the database in Figure 5.5 in SQL. Show the query results if each query is applied to the database state in Figure 5.6.

a. For each department whose average employee salary is more than $30,000, retrieve the department name and the number of employees working for that department.

b. Suppose that we want the number of male employees in each department making more than $30,000, rather than all employees (as in Exercise 7.5a). Can we specify this query in SQL? Why or why not?

a.
```
select Dname, count(*)
from DEPARTMENT D, EMPLOYEE E
where D.Dnumber = E.Dno
group by Dname
having avg(E.Salary) > 30000;
```

Dname	count(*)
Administration	3

Headquarters	1
Research	4

b. 可以。只要先選出男性且薪水高於 30000 的員工，再計算每個部門的員工數量即可。

```
select Dname, count(*)
from DEPARTMENT D, EMPLOYEE E
where D.Dnumber = E.Dno and E.Sex = 'M' and E.Salary > 30000
group by Dname;
```

Dname	count(*)
Headquarters	1
Research	2